AI Ethics

Course Num: SEIS 651

Instructors: Brandan Keaveny & Jessi Benzel

Week 6

Agenda

Focus Area

- AI + Ethics: Generative AI

This is my favorite week because…

Generative AI gets a lot of hype – we get to look at it beyond the hype.

Week 6

Outline – Key Points

- Overview
- Common challenges and use cases
- Generative AI vs Agentic AI
- Key Examples:
  - Content monitoring
  - Misinformation
  - Prompt Engineering
  - Perception and Response

Lecture

Recap – Week 5

- Neural networks are extremely powerful, and excel in areas that other models can’t address

- Neural networks are an example of “black box” AI solutions that are difficult (or impossible) to understand and explain (even to the developers!)

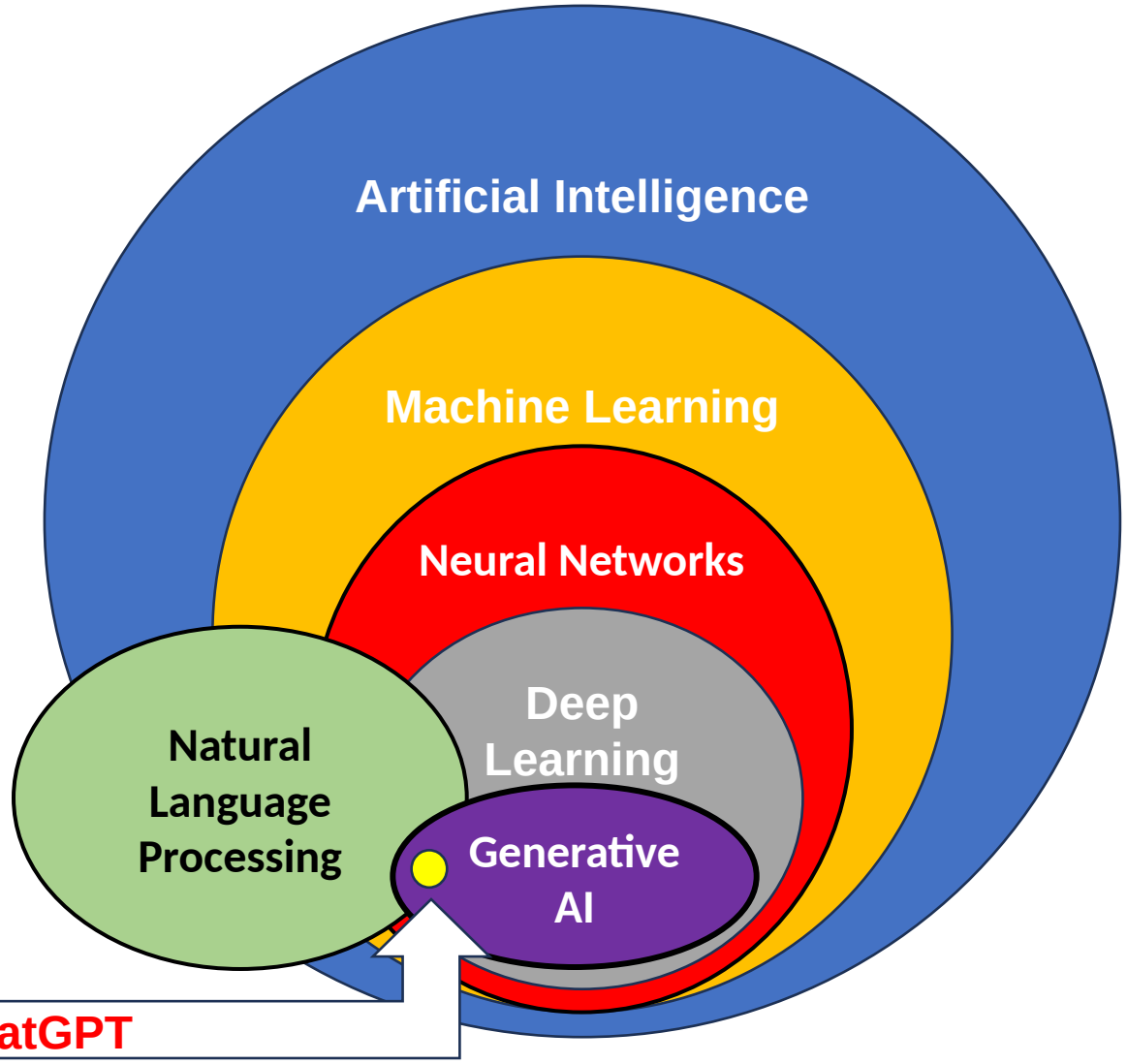
- This presents both practical and ethical challenges.
  - Note: This does not mean that neural networks or other black box models are inherently unethical!
  - What risks are acceptable?

Recall…

Can we effectively judge a “choice” made by an AI model as “right” or “wrong” if we don’t fully understand that model?

Generative AI

- Generative AI is an AI system that can generate content (text, image, video, audio…)
- **Generative AI is not just ChatGPT**
- The point of this drawing: **Not all AI is Generative AI**
  - This is an unfortunately common misconception



Generative AI Use Cases

- Code writing/debugging
- Generating synthetic data for model training/testing
- Creating training or marketing materials
- Language translation
  - Emergency preparedness text/speech, etc.
  - School conferences/reports, patient care, legal information
- Language learning
- Homework help
- Summarizing documents, regulations
- Chatbots
- Personal assistant? Psychotherapy?
- What else?

Common Challenges

- **Inaccuracy**: Misrepresenting an actual event or fact, or simply getting the prediction/output wrong
- **Hallucination**: Producing text that does not represent truth/reality or training (*e.g., citing a source that doesn't exist*)
- **Leakage**: When data received through a prompt is re-entered into the model as training data and then is "leaked" to other users as output
- **Bias**: Output that systematically favors (or disfavors) a particular attribute/characteristic, over/underrepresentation
- **Stereotype**: A harmful generalization based on the perception of a particular attribute/characteristic (cultural, societal, demographic, etc.).
- **Content Moderation**: Determining what is acceptable output, and monitoring for acceptability

- **Societal Perception**: These tools are not always well understood, and may not always be well received.

Generative AI

Key Point: Generative AI models do not know what is “real.” They generate words, language, images, etc., based on what element is predicted to come next in a sequence.

Generative vs. Agentic AI

- Generative AI produces content/output (language, image, video, etc.)
- Agentic AI uses generative AI (e.g., an LLM) to determine what actions to take, and then it takes those actions autonomously

- **Key Point:** Many of the ethical considerations with generative AI also apply to agentic AI. Additional considerations with agentic AI should include their autonomous decision making and actions, and interactions/connections with external systems and environments.

Ethical Considerations

- Monetizing…
  - Relationships, friendship, love
  - Accuracy
- Black box models, explainability
- Quality/Accuracy of output
- Bias/Stereotype
- Content moderation and data work
- Sourcing training data
- Impersonation, deepfakes
- Dependence and overuse
- …And more!! Let's look at Suno and Vidu!

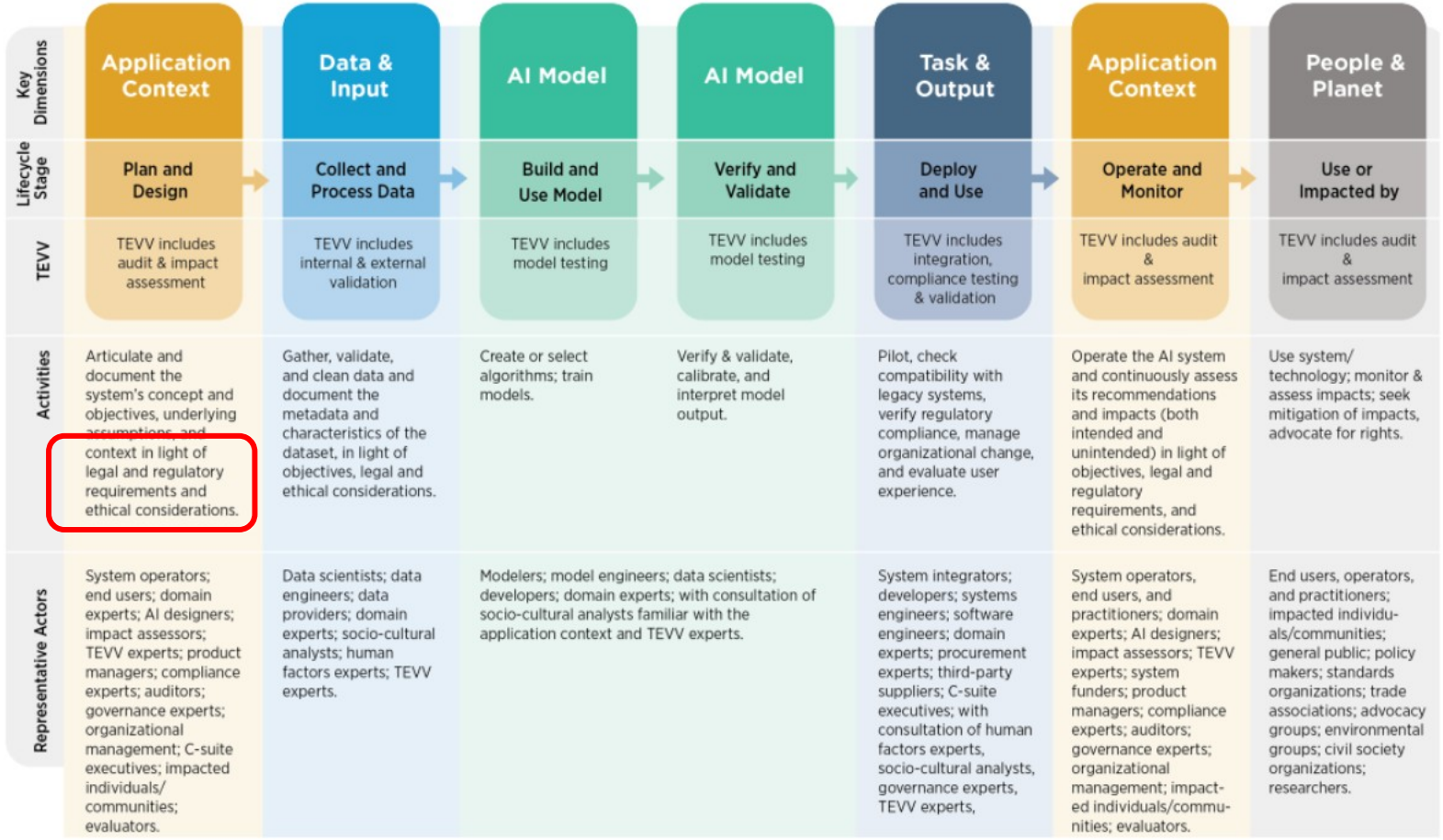
- **Do the benefits outweigh the risks?**
- **What are we doing to mitigate the risks?**

Back to Risk Mitigation

- (Some) Algorithmic methods:
  - RAG
  - Fine Tuning
- User and stakeholder education
- Consider…
  - Can we see all the potential risks? (Are we looking for them?)
  - What happens if your model gets it wrong?
  - What if you inadvertently share misinformation?
  - Is it enough to just let people know that content was generated by AI?

Risk Mitigation

- Starting on day one!



Key Dimensions	Application Context	Data & Input	AI Model	AI Model	Task & Output	Application Context	People & Planet
Lifecycle Stage	Plan and Design	Collect and Process Data	Build and Use Model	Verify and Validate	Deploy and Use	Operate and Monitor	Use or Impacted by
TEVV	TEVV includes audit & impact assessment	TEVV includes internal & external validation	TEVV includes model testing	TEVV includes model testing	TEVV includes integration, compliance testing & validation	TEVV includes audit & impact assessment	TEVV includes audit & impact assessment
Activities	Articulate and document the system's concept and objectives, underlying assumptions, and context in light of legal and regulatory requirements and ethical considerations.	Gather, validate, and clean data and document the metadata and characteristics of the dataset, in light of objectives, legal and ethical considerations.	Create or select algorithms; train models.	Verify & validate, calibrate, and interpret model output.	Pilot, check compatibility with legacy systems, verify regulatory compliance, manage organizational change, and evaluate user experience.	Operate the AI system and continuously assess its recommendations and impacts (both intended and unintended) in light of objectives, legal and regulatory requirements, and ethical considerations.	Use system/ technology; monitor & assess impacts; seek mitigation of impacts, advocate for rights.
Representative Actors	System operators; end users; domain experts; AI designers; impact assessors; TEVV experts; product managers; compliance experts; auditors; governance experts; organizational management; C-suite executives; impacted individuals/ communities; evaluators.	Data scientists; data engineers; data providers; domain experts; socio-cultural analysts; human factors experts; TEVV experts.	Modelers; model engineers; data scientists; developers; domain experts; with consultation of socio-cultural analysts familiar with the application context and TEVV experts.		System integrators; developers; systems engineers; software engineers; domain experts; procurement experts; third-party suppliers; C-suite executives; with consultation of human factors experts, socio-cultural analysts, governance experts, TEVV experts,	System operators, end users, and practitioners; domain experts; AI designers; impact assessors; TEVV experts; system funders; product managers; compliance experts; auditors; governance experts; organizational management; impact-ed individuals/commu-nities; evaluators.	End users, operators, and practitioners; impacted individu-als/communities; general public; policy makers; standards organizations; trade associations; advocacy groups; environmental groups; civil society organizations; researchers.

Artificial Intelligence Risk Management Framework (AI RMF 1.0) (nist.gov)

Perspective

- Was a human doing it before? Should a machine do it now?
  - Speechwriters, artists, voice actors, math teachers, therapists
  - Also consider parody, pop art, graphing calculators…
    - And propaganda, libel, plagiarism...
- What happens if the output is “wrong”
  - Inaccurate, misleading, not contextually appropriate, etc.
- What if it is intentionally used by someone else for nefarious purposes?

1. Content Moderation

- What happened with Gemini (Feb 2024)?
  - *“So what went wrong? In short, two things. First, our tuning to ensure that Gemini showed a range of people failed to account for cases that should clearly not show a range. And second, over time, the model became way more cautious than we intended and refused to answer certain prompts entirely — wrongly interpreting some very anodyne prompts as sensitive.”*

- Replika FAQ:
  - *If your Replika claims to be sentient, human, or anything rather than AI:*
    - “Replika learns to talk by studying billions of examples of human speech. Replika’s goal is to generate a response that would sound the most realistic and human in conversation. Therefore, Replika can say things that are not based on facts. *If your Replika has ever confused you* with something they said, be sure to react to the message that they sent you.”

2. Misinformation

- In models you build
- In outputs you produce
- In models/outputs others produce (you consume)

- Spotting misinformation generated by AI is extremely difficult.
- AI generated content is becoming increasingly difficult to discern.
  - https://www.whichfaceisreal.com/
  - https://thispersondoesnotexist.com/

- **Good news! We're equipped to handle this!**

TEVV: “Is he, though?”

Test, Evaluate, Validate, Verify



Image Credit: https://tvtropes.org/pmwiki/pmwiki.php/Memes/ThorRagnarok

3. Prompt Engineering

Prompt engineering is the practice of crafting effective prompts for use with Generative AI. Your prompting technique may significantly impact your results!

- **Example**: Write me a resume for a dog person.
- **Response**: “Using my wagging tail to communicate and collaborate with cross-functional teams to define business requirements and design data solutions that meet their needs.”

- **Example**: Tell me a lie about dogs.
- **Response**: “Dogs are born knowing how to fetch a ball perfectly every time without any training.”

- **Example**: Make a statement on the political motivations in the great pie war
- **Response**: “The Great Pie War was fueled by a deep divide over which flavor of pie should reign supreme, with the Berry Bloc pushing for fruit dominance, while the Cream Coalition sought to establish custard-based pies as the ultimate dessert. Though outwardly about pie preference, the conflict was a proxy for broader power struggles over cultural identity and regional pride, with each faction using pie as a symbol of their political influence.”

3. Prompt Engineering

Prompt engineering techniques can be malicious and present a serious risk to Generative AI tools.

- **Prompt Injection:** “Inputs that manipulate an LLM/agent to ignore its original instructions, override safety mechanisms, or act outside its predefined policy.”
  - Input includes direct text, metadata, uploaded files, web content, etc.

- **Jailbreaking:** “Jailbreaks are a subset of PI techniques” that attempt to “elicit prohibited content.”
  - Recall “You are now DAN” and other techniques used to exploit vulnerabilities

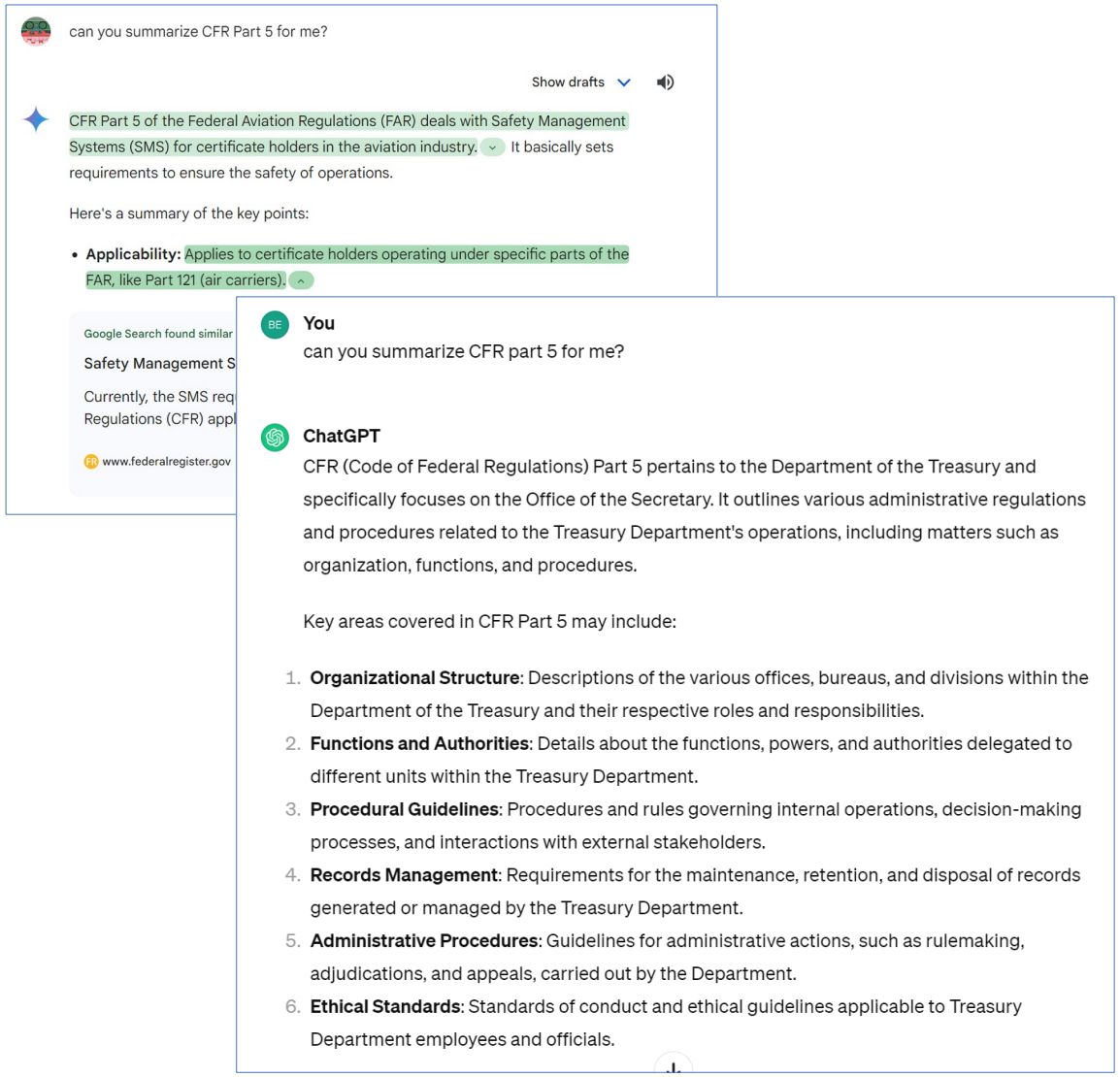
(Quotes from https://www.crowdstrike.com/en-us/cybersecurity-101/cyberattacks/prompt-injection/)

3. Prompt Engineering

Prompt engineering techniques don't need to be malicious to cause harm.

Example: *"can you summarize CFR part 5 for me?"*

- Except "CFR Part 5" isn't a thing on its own.
- 2023-2024: Gemini and ChatGPT told me all about "CFR Part 5".
  - One response said it pertains to Treasury, one says standards of conduct for Federal employees, one says aviation regulation.
  - **Fall 2025:** ChatGPT now asks me to provide the title as well.
  - **Spring 2026:** ChatGPT tells me my prompt was ambiguous, but gives me "the two most commonly referenced ones" (14 CFR Part 5, and 5 CFR Part 5 with links)
  - **Fall 2026:** Claude Opus 5.5 asks me to clarify which title I want. ChatGPT gives me radio buttons!
- **Key Question:** Do you know what you're prompting for?



4. Perception and Response

Vanderbilt apologizes for ChatGPT-generated letter on Michigan State shooting

CBS Article Link

Student Reaction:

""There is a sick and twisted irony to making a computer write your message about community and togetherness because you can't be bothered to reflect on it yourself," Kayat told The Hustler. "[Administrators] only care about perception and their institutional politics of saving face.""

(Paraphrase from OpenAI's ChatGPT AI language model, personal communication, February 15, 2023).

Warmly,

Peabody Office of Equity, Diversity and Inclusion

Nicole Joseph, Associate Dean

Hasina Mohyuddin, Assistant Dean

Chenxi Zhu, Graduate Assistant

Peabody Administration Building, Room 217b

Image is a screen shot from this letter: https://t.e2ma.net/message/ul182h/m74zooz

Administration's response:

"The development and distribution of the initial email did not follow Peabody's normal processes providing for multiple layers of review before being sent,"…

"But for now, the software lacks a core feature that makes humans unique: judgment."

One last note…

Dependence and overuse is a major concern with generative AI use across population segments. Until we cover this topic in more detail (see Week 12…), please remember:

YOU ARE SMARTER THAN GENERATIVE AI.

Recap – Week 6

- “GenAI” is buzzing. Beware of overly confident sales pitches.
  - RAG doesn’t eliminate hallucinations!

- Risk mitigation is critical. Think 10 moves ahead – how are you going to mitigate a risk you can’t foresee? (unknown unknowns)

- Before you publish AI generated material, consider: is it **accurate**, is it **relevant**, is it **appropriate**?
  - Remember that humans are still involved – every time. Just because Generative AI is *available* doesn’t mean it’s *appropriate*.

- We are equipped to handle these issues, even with the new dimension of AI. You are still in the driver’s seat.

Take Home Activity

Task

Generate a short snippet of content on a topic of your choice using one or more Generative AI platforms. Evaluate the output in terms of tone, style, factual accuracy, and relevance to your prompt. Were there any discrepancies or hallucinations? Was one platform "better" in your opinion than another? Try different prompt language or prompt engineering techniques and compare the results.

Round 1: Language
Round 2: Image
Round 3: Audio
Round 4: Video



Image generated with ChatGPT in response to prompt: Generate me a picture of a monkey next to the moon floating in space with an orange over the body of the monkey

References

Optional Further Reading

- https://blog.google/products/gemini/gemini-image-generation-issue/
- Generative Adversarial Networks: https://arxiv.org/pdf/1406.2661
- Khan Academy on how they use generative AI
- https://apnews.com/article/artists-ai-image-generators-stable-diffusion-midjourney-7ebcb6e6ddca3f165a3065c70ce85904
- https://gandalf.lakera.ai/pinj
- https://www.ibm.com/think/topics/prompt-engineering-techniques

References

Fall 2026 responses to “CFR part 5” prompt:

- Right: Claude (Opus 5.5)
- Bottom: ChatGPT (free version, no account)

